University of Eloued
Faculty of Exact Sciences
Department of Computer science
For Second-year Master: IA & Data Science

University season: 2026/2027
Module:Big Data Analytics

TP02: How to Work with Big Data Files (+5GB)

All work is uploaded to your GitHub account.

1. Educational Objectives

- Understand the Out-Of-Memory (OOM) limitations of traditional data loading methods.
- Implement Chunking techniques using standard Pandas workflows.
- Leverage Parallel and Lazy Computing with Dask DataFrames.
- Benchmark and profile performance metrics (Execution Time vs. Memory Footprint).

2. Prerequisites & Environment Setup

Students must install the following core Python libraries before starting the lab:

pip install pandas dask[complete] psutil matplotlib pyarrow

3. Guided Lab Tasks

Task 1: The Traditional Failure (OOM Challenge)

Attempt to read the 5GB+ dataset using standard pd.read_csv(). Monitor system behavior via psutil or task manager, and document the MemoryError exception.

Task 2: Stream Processing (Pandas Chunking)

Process the large dataset in chunks of 100,000 rows. Compute incremental aggregation metrics (e.g., global mean or sum) without overloading the system memory.

Task 3: Parallelized Out-of-Core Compute (Dask Framework)

Load the data using dask.dataframe. Explain the concept of Lazy Evaluation, optimize the compute graph, and compare the execution structure.

4. Expected Performance Benchmark

Method	Max RAM Utilization	Execution Speed
Standard Pandas	100% (Crash)	N/A (OOM Error)
Pandas Chunking	Low (~Constant)	Moderate (Sequential)
Dask DataFrame	Low-Optimized	Fast (Multi-threaded)

Good luck